Computer Vision – Assignment 5

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Segmentation using background subtraction

```python
#Loading of the video
video_path = 'static_camera.webm'
cap = cv2.VideoCapture(video_path)

#It reads the first frame to use it
ret, frame = cap.read()
if not ret:
    raise Exception("Can't read the video.")

gray_frame = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
height, width = gray_frame.shape
num_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
```

The first necessary thing is the loading of the video and the read of the first frame.

```
21  #Accumulated Weighted Image
22  accum_weight = 0.01
23  accum_background = np.float32(gray_frame)
24
25  #MOG2 Background Subtractor
26  mog2 = cv2.createBackgroundSubtractorMOG2()
27
28  #Custom Moving Average Background
29  custom_background = np.zeros_like(gray_frame, dtype=np.float32)
30
31  #Store foregrounds for the chosen frame
32  chosen_frame_index = num_frames // 2
33  foregrounds = {}
```

The set of the Accumulated Weighted Image, the backgrounds and the foregrounds.

```
#Process video
cap.set(cv2.CAP_PROP_POS_FRAMES, 0)  # Restart video

for i in range(num_frames):
    ret, frame = cap.read()
    if not ret:
        break

    gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)

    #Accumulate Weighted
    cv2.accumulateWeighted(gray, accum_background, accum_weight)
    fg1 = cv2.absdiff(gray, cv2.convertScaleAbs(accum_background))

    #MOG2
    fg2 = mog2.apply(frame)

    #Custom Moving Average
    custom_background = (custom_background * i + gray) / (i + 1)
    fg3 = cv2.absdiff(gray, cv2.convertScaleAbs(custom_background))

    #Save foregrounds at the chosen frame
    if i == chosen_frame_index:
        foregrounds['frame'] = frame
        foregrounds['fg1'] = fg1
        foregrounds['fg2'] = fg2
        foregrounds['fg3'] = fg3
```

During the process of the video, we set the functions for the Accumulated Weight, the MOG2 and the Moving Average. Also, it is really important to set and save the foregrounds.

```python
#Final background models
final_bg1 = cv2.convertScaleAbs(accum_background)
final_bg2 = mog2.getBackgroundImage()
final_bg3 = cv2.convertScaleAbs(custom_background)

cap.release()

#Visualization
def show_image(title, img, cmap='gray'):
    plt.figure(figsize=(4, 4))
    plt.title(title)
    plt.imshow(img, cmap=cmap)
    plt.axis('off')

#Background
show_image("Background - Accumulate Weighted", final_bg1)
show_image("Background - MOG2", final_bg2)
show_image("Background - Moving Average", final_bg3)

#Original Frame and Foreground masks
show_image("Original Frame", cv2.cvtColor(foregrounds['frame'], cv2.COLOR_BGR2RGB), cmap=None)
show_image("Foreground - Accumulate Weighted", foregrounds['fg1'])
show_image("Foreground - MOG2", foregrounds['fg2'])
show_image("Foreground - Moving Average", foregrounds['fg3'])

plt.show()
```

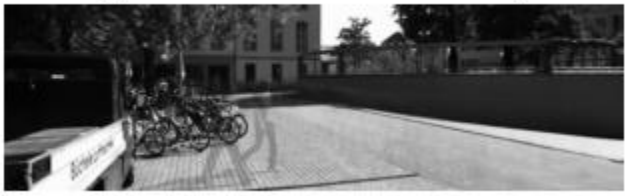
The backgrounds are set and the results are printed.

Results

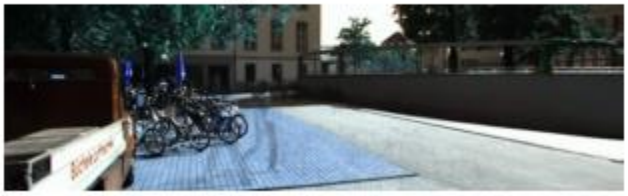
Foreground - MOG2

Foreground - Moving Average

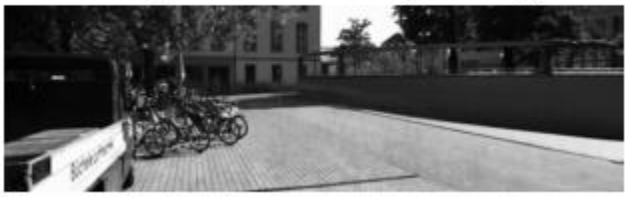
Background - AccumulateWeighted



Background - MOG2



Background - Moving Average



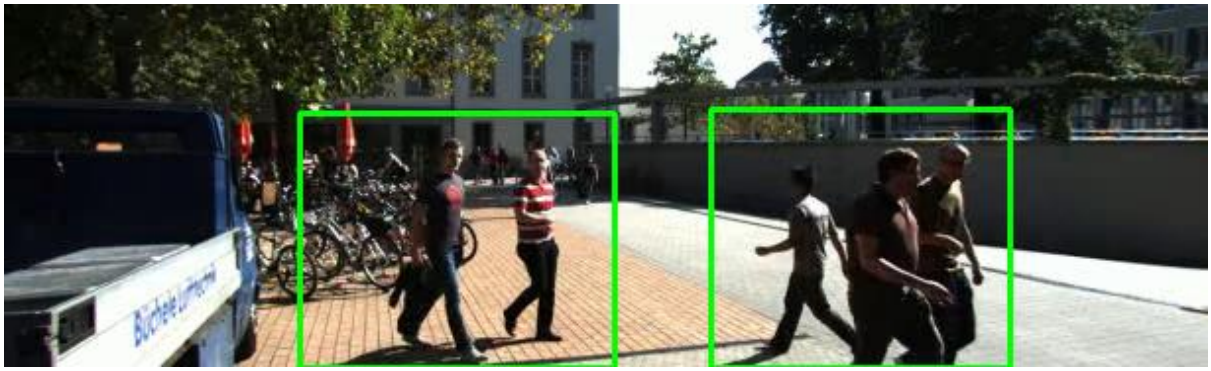
These are the results that we get for the Foregrounds and the Backgrounds from the implementation of the Accumulated Weighted, MOG2 and Moving Average.

Dense Optical Flow

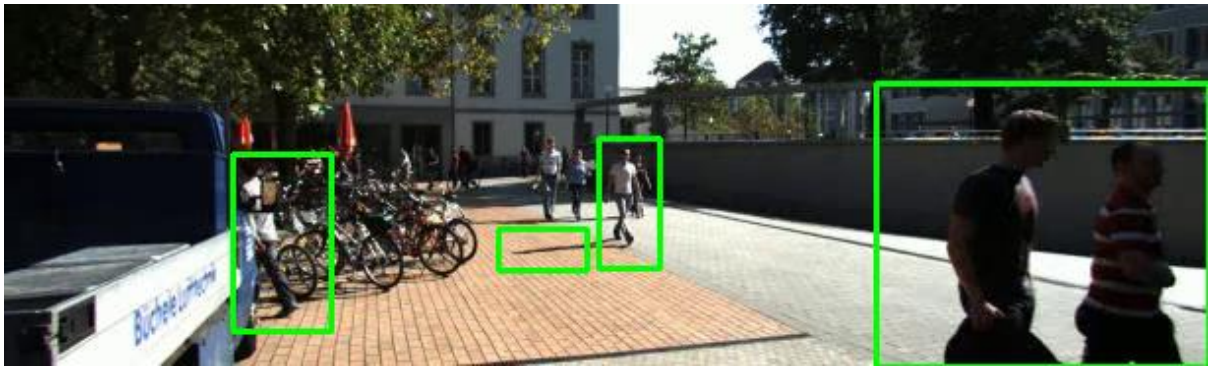
Results

Frame 2

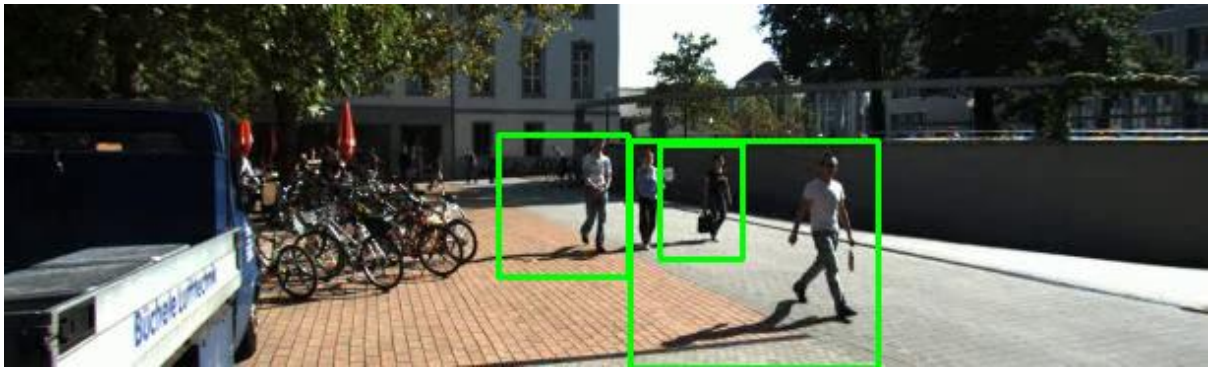
Frame 32



Frame 62



Frame 92



Frame 122